

Advanced Econometrics

1 Unordered choice models

Exercise 1

The dataset consists of 1182 observations on individuals' choices of how to fish. There are four fishing modes (beach, pier, boat, charter), two alternative specific variables (price and catch), and one choice/individual specific variable (income)¹.

- Estimate a pure multinomial model for `mode`.
- Determine and interpret marginal effects.
- Verify the Independence from Irrelevant Alternatives (IIA) assumption.
- Check whether alternatives *boat* and *charter* might be combined into one category.
- Test joint insignificance of all variables in the model.
- Interpret pseudo- R^2 determined by `mlogit` function.

Exercise 2

We observe 1822 purchases, covering 104 weeks and 5 stores, in which a consumer purchased 2-liter bottles of either Pepsi (34.6%), 7-Up (37.4%), or Coke Classic (28%). These data are in the file `cola.csv`. In the sample, the average price of Pepsi was \$1.23, of 7-Up \$1.12, and of Coke \$1.21. We estimate the conditional logit model that choices depend on *price*, *feature*, and *display* variables.²

variable	variable label
id	customer id
pepsi	= 1 if pepsi chosen, otherwise = 0
sevenup	= 1 if 7up chosen, otherwise = 0
coke	= 1 if coke chosen, otherwise = 0
pr_pepsi	price of 2 liter bottle of pepsi
pr_7up	price of 2 liter bottle of 7up
pr_coke	price of 2 liter bottle of coke
feat_pepsi	= 1 if pepsi is a featured item at the time of purchase, otherwise = 0
feat_7up	= 1 if 7up is a featured item at the time of purchase, otherwise = 0
feat_coke	= 1 if coke is a featured item at the time of purchase, otherwise = 0
disp_pepsi	= 1 if pepsi is displayed at time of purchase, otherwise = 0
disp_7up	= 1 if 7up is displayed at time of purchase, otherwise = 0
disp_coke	= 1 if coke is displayed at time of purchase, otherwise = 0

- Why shouldn't we use a multinomial logit model for this problem?

¹Estimation of multinomial logit models in R: The mlogit Packages, Yves Croissant, A Vignette for `mlogit` Package.

²Principles of Econometrics, Third Edition, by R. Carter Hill, William E. Griffiths and Guay C. Lim, page 433.

- b) Estimate a conditional logit model for soda choice with covariates mentioned above.
- c) Determine and interpret pseudo- R^2 statistics.
- d) Are variables in the model jointly significant?
- e) Determine and interpret marginal effects for *price*.
- f) Determine and interpret marginal effects for *display*.

Exercise 3

For the dataset in exercise 1 estimate the "impure" conditional logit model. Assume that fishing mode depends not only on *income* but also on *price* and *catch* rate.

- a) Why this model is "impure"?
- b) Estimate the model.
- c) Perform goodness-of-fit tests.
- d) Check the joint significance of the variables in the model.
- e) Which variables have an impact on the choice of fishing mode?
- f) Determine marginal effects and interpret their values.

Exercise 4

Occupations might be organized as professional, white collar, blue collar, craft, and menial, which is the example we use in this chapter.³ The `nomocc2.csv` dataset comprises, beside job types (*occ*), race (*white* – 1=white 0=nonwhite), years of education (*ed*), and years of work experience (*exper*).

- a) What model should be used for *occ*?
- b) Estimate appropriate model.
- c) Can we combine alternatives *bluecollar* with *craft*?
- d) Does *bluecollar* alternative violates IIA assumption?
- e) Determine marginal effects for *exper* and interpret their values.

Exercise 5

We have data on 295 consumers and their choice of automobile. Each consumer chose among an American, Japanese, or European car; the variable *car* indicates the nationality of the car for each alternative. We want to explore the relationship between the choice of car to the consumer's sex (variable *sex*) and income (variable *income* in thousands of dollars). We also have information on the number of dealerships of each nationality in the consumer's city in the variable *dealer* that we want to include as a regressor. We assume that consumers' preferences are influenced by the number of dealerships in an area but

³Regression Models for Categorical Dependent Variables Using Stata, Scott J. Long, Jeremy Freese, Stata Press. The dataset in the exercise accompanies the book.

that the number of dealerships is not influenced by consumer preferences (which we admit is a rather strong assumption). Each consumer's chosen car is indicated by the variable *choice*.⁴

1. Import the data from `car_choice.csv` file.
2. Is the data already in the long form?
3. Which variables are alternative-specific? Which are case-specific?
4. Why shouldn't we use ordered logit for this problem?
5. Estimate the model.
6. Calculate marginal effects and interpret them.

Exercise 6

Read in the data stored in the file *fml142_part.csv*. This dataset comes from Consumer Expenditure Surveys by Bureau of Labor Statistics of the United States Department of Labor. We are interested in estimating the model for employment type variable (*empl_yp1*) with respect to age (*age_ref*), sex (*sex_ref*), family size (*fam_size*), and marital status (*marital1*).

- a) Estimate the multinomial logit model with `multinom` function of `nnet` library.
- b) Estimate the multinomial logit model with `mlogit` function of `mlogit` library.

⁴Stata documentation on `asclogit` function, p. 90-91.